

Ong Chang Heng Bertrand

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EDUCATION

National University of Singapore (NUS)

Bachelor of Computing (Honours) in Computer Science

Singapore

Aug 2022 – May 2026

- Grade: Second Class (Upper) Honours
- Focus Area: Software Engineering & Database Systems (Distinction)
- NUS Overseas College (NOC Shanghai) Program, Entrepreneurial Studies

WORK EXPERIENCE

Funding Societies

Software Engineer Intern | Typescript, Python

Singapore

Jan 2026 – Present

- Engineering workflow automation systems to streamline internal processes, improving team productivity and saving up to 200 hours across various departments per week.
- Contributing to the open-source project Activepieces, implementing production-grade features and maintaining code quality through reviews and testing.
- Developing AI gateways to monitor and optimize LLM and external API usage costs, improving observability and cost control across services.

DeepVerse

Backend Engineer Intern | Python, PostgreSQL, MongoDB, Docker

Shanghai, China

Aug 2024 – Jul 2025

- Designed, developed, and tested scalable RESTful APIs using Python FastAPI framework, interfacing with PostgreSQL and MongoDB to support algorithm and frontend requirements.
- Engineered data transformation pipelines that converted algorithm training outputs into structured graph data to be consumed by frontend dashboards.
- Set up and deployed gRPC-based microservices to modularize communication between algorithmic models and database layers, improving system scalability and latency.
- Integrated DeepSeek LLM into a Chat module; Optimized Retrieval-Augmented Generation (RAG) latency, reducing response time by 50%.
- Implemented structure-aware text chunking with LangChain, preserving semantic context around tables, equations, and text to improve document parsing accuracy.

Boxgreen

Software Engineer Intern | Shopify, Javascript

Singapore

May 2024 – Aug 2024

- Maintained and developed new functionalities on Boxgreen's website, resulting in a 20% increase in platform traffic.
- Developed an e-commerce store for its partner store, The Good Market, achieving more than 100 sales upon launch.
- Designed and built an internal Objectives and Key Results (OKR) web application to support employee performance reviews and reporting workflows.

PROJECTS

Peerprep | React, Typescript, Python, Node, Kafka, Docker, PostgreSQL, Redis

Aug 2025 – Nov 2025

- Built a collaborative coding web application enabling peer matching and synchronized problem-solving sessions.
- Designed and implemented an event-driven microservices architecture using Kafka, containerized with Docker, and deployed on AWS EC2.
- Implemented low-latency collaboration services using web sockets, leveraging Redis for in-memory state management and caching.

SideQuest | Golang, React, Supabase

Dec 2025 – Present

- Developing a full-stack productivity application that converts long-term goals into actionable weekly tasks.
- Implemented backend services in Go, with a React frontend and Supabase for authentication and database management.

CO-CURRICULAR

NUS Squash | Captain

Apr 2025 – Present

- Led a competitive team to 1st place at the Singapore University Games (SUniG) AY 25/26.
- Directed end-to-end team operations, including training program design, logistics coordination, and financial management.
- Organized 'Squash For All', a community outreach initiative introducing squash to children and first-time players.

TECHNICAL SKILLS

Languages: Java, Javascript, Typescript, Python, C, Golang

Technologies: FastAPI, PostgreSQL, MongoDB, React, Redis, Git, Docker, Node, Express, Kafka, Linux